
Drift $\neq$ Damage: A Counterfactual Test of Encoder Adaptation in Time-Series Foundation Models

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Abstract

1 Fine-tuning a time-series foundation model (TSFM) substantially changes its
2 representations, and preservation methods exist to limit that change. But does the
3 change tell us whether *allowing* encoder adaptation was useful? We test that directly
4 with a **frozen-encoder counterfactual**: full fine-tuning (B) against encoder-frozen
5 adaptation (D), their difference B–D gated on a **pretrained-capability inclusion**
6 **criterion**. Our headline finding is **methodological: B–D must be scored on**
7 **held-out windows**. Read on the windows model selection already uses, it flatters the
8 frozen encoder: **selection-window bias**. Re-scored on a chronologically disjoint
9 test split, **four of 31 cells reverse sign** (Chronos/ETTh1 $+6.8 \rightarrow -39.2$ pp),
10 every one in the *same* direction. Second, **within a backbone we find no reliable**
11 **evidence that CKA magnitude predicts the held-out incremental value of**
12 **allowing encoder-side adaptation** under our evaluated protocols; across three
13 architecturally distinct TSFMs (statistically within Moirai, directionally elsewhere),
14 two Moirai-Large $h=96$ cells 0.004 apart in CKA differ by 33 pp. Third, the
15 intervention exposes a regime **within Moirai** we name **pretrained-capability**
16 **degradation**: in eight cells meeting the criterion, full fine-tuning leaves the model
17 **worse than not fine-tuning at all**, in every seed, while a frozen encoder *improves*
18 it. **But the criterion does not predict where**: pre-registered on eight new cells it
19 flagged all eight and only two degraded. We establish this regime’s **existence, not**
20 **its prevalence**: the third backbone contains no such cell.

21 1 Introduction

22 Fine-tuning changes representations. That observation alone cannot tell us whether the change was
23 useful. We therefore compare full fine-tuning against an encoder-frozen counterfactual and score the
24 difference on *held-out* data.

25 A Moirai encoder fine-tuned on ETTh2 can have CKA with its pretrained self fall to ~ 0.3 (a $\sim 70\%$
26 representational shift), and yet forecast *better* on the very same task. If drift were a proxy for damage,
27 this should not happen.

28 Preservation methods such as LoRA, EWC and L2-SP [Kirkpatrick et al., 2017, Hu et al., 2022, Li
29 et al., 2018] exist to limit how far fine-tuning moves a released checkpoint from its pretrained state.
30 The **reflex** is to reach for them whenever drift is large, without first asking whether the pretrained state
31 was worth preserving here. This bundles three implicit claims (drift implies forgetting, forgetting
32 implies damage, drift should be minimized), and the bundle comes apart on TSFMs. Our “ $\neq$ ” is
33 insufficiency, not independence: drift does not *reliably imply* damage, because **downstream utility**
34 **depends on the functional role of the changed representation**, not *how much* a backbone drifts but
35 *what kind* of change it is, which one scalar cannot distinguish.

Three quantities, three questions, and the paper turns on not conflating them.

representation drift (CKA)	how much changed?	observational
B–D	was allowing the change worth it?	interventional
forecast MSE	did the model improve?	outcome

CKA measures the first, B–D the second, neither is the third; *damage* is what the reflex infers, not what we measure.

We contribute a within-model dissociation (Fig. 2, App. A), a capability check across backbones with controlled encoder interventions on the cells that pass it, and a diagnostic chain needing only released weights and target data (Fig. 1). **What is new is the combination:** a frozen-encoder counterfactual gated on *demonstrated* pretrained capability, scored on held-out windows, and run on released checkpoints. On scope, **our strongest evidence comes from Moirai:** Chronos and TimesFM provide directional cross-architecture evidence, while MOMENT contributes no intervention evidence, its runs being incomplete.

Our claims, strongest evidence first. (1) **B–D must be scored on held-out windows:** read on the windows model selection uses, it flatters the frozen encoder, four cells reverse sign, and the gate licensing them is vulnerable the same way. (2) **Within a backbone we find no reliable evidence that CKA magnitude predicts the held-out incremental value of allowing encoder-side adaptation** under our evaluated protocols. (3) **The intervention exposes pretrained-capability degradation:** eight Moirai cells in which full fine-tuning ends up worse than not fine-tuning at all while a frozen encoder improves on it. The downstream-only criterion that marks where capability is demonstrated is *not* a validated detector of that regime, as our pre-registered arm shows.

2 Setup and the Diagnostic Framework

We study four TSFMs: **Moirai** [Woo et al., 2024] (encoder-only, MoE), **Chronos-T5** [Ansari et al., 2024] (encoder-decoder), **TimesFM** [Das et al., 2024] (decoder-only, $d_{\text{model}}=1280$), and **MOMENT** [Goswami et al., 2024] (encoder-only). Data is univariate ETT [Zhou et al., 2021]/M4 (lookback 96; $h=24$ cross-backbone, $h=96/192$ Moirai sweep), plus ILI (weekly influenza, 7 features, lookback 104, $h=24$) and Electricity, *restricted to 7 of its 370 series* (App. E). **Each arm trains through its own objective and head;** only the 5 Chronos cells attach a common MSE head to encoder states, hence limitation (iii), and MSE magnitudes are *not* comparable across arms, so B–D too is read within a backbone (App. E). Two conditions run throughout: **B** (full fine-tuning) and **D** (frozen encoder, head-only). Forgetting% = $100(\text{MSE}_{\text{FT}} - \text{MSE}_{\text{ZS}})/\text{MSE}_{\text{ZS}}$; negative means improvement over zero-shot.

Pretrained-capability inclusion criterion. Before asking what adaptation is worth, we ask whether there is capability worth preserving, not whether fine-tuning is worthwhile. The check needs only released weights and the practitioner’s series: does the released model beat a lookback-96 linear baseline by >20%, that is, $R_{\text{task}}^2 = 1 - \text{MSE}_{\text{ZS}}/\text{MSE}_{\text{Linear}} \geq 0.20$? **Its role is inclusion, not prescription: it decides which cells make an intervention interpretable, and is not a validated deployment-time decision rule.**

CKA drift. Linear CKA [Kornblith et al., 2019] between pretrained and fine-tuned backbone on mean-pooled encoder outputs over identical held-out 192-timestep windows; flagged at < 0.95 and compared only *within* a backbone, never across architectures. CKA remains a valid similarity measure; our claim is only that its magnitude should not be read as downstream damage. Pooling moves the CKA magnitude but not the flag (App. M, I).

Frozen-encoder intervention (B–D). Condition D measures what head-only training achieves when the encoder cannot move; B–D is the resulting forgetting% gap in percentage points, negative meaning adaptation helps. Freezing applies to encoder weights only (input/output projections stay trainable), so D is not representation-frozen (CKA 0.38–1.000). **B–D therefore estimates the incremental downstream value of allowing encoder-side adaptation under this training protocol, not the causal effect of the representation change CKA measures.** Freezing also removes optimization degrees of freedom, so no single mechanism is isolated by this contrast.

3 Results: Three Findings

Table 1 summarizes the cross-backbone results, ordered by B–D; pretrained value proves to be a property of checkpoint $\times$ dataset $\times$ horizon rather than of an architecture (App. E).

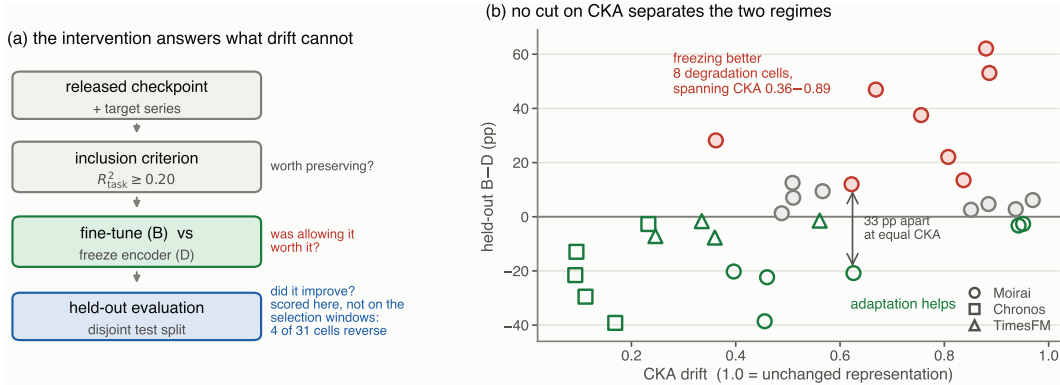


Figure 1: **Drift cannot stand in for the intervention.** (a) The last step must be asked on held-out windows: scored on the selection split instead, four of 31 cells come out with the *opposite* sign, every one flattering the frozen encoder. (b) All 31 cells; red marks the eight **pretrained-capability degradation** cells, and the two Moirai-Large $h=96$ cells at CKA 0.622/0.626 fall on opposite sides of the regime; the x -axis is not a decision boundary (App. F).

Finding 1 (methodological): B–D must be scored on held-out windows. Two splits, used throughout: the **selection split** trains the models and chooses among them, and the **test split** is chronologically disjoint and used for neither. **B–D read on the selection split estimates no generalization quantity, and its sign is not safe there: selection-window bias.** The reason is that B–D compares two trained models and inherits the model selection of both: B early-stops on the selection split and a frozen head’s regularizer is tuned there, so both arms are tuned on the windows the difference is then read on, and nothing constrains them to profit from that equally. Re-scored on the test split, **four of the 31 cells reverse sign** (Chronos/ETTh1 largest at $+6.8 \pm 2.9 \rightarrow -39.2 \pm 13.9$, joined by both Moirai/ETTh2 cells and TimesFM/ETTh2, whose $+0.2 \pm 2.9$ leaves it the least informative of the four), while others merely soften, and all four run the same way: on the selection split, B–D *flatters the frozen encoder*. We cannot separate selection overfitting from validation-to-test shift (the *untrained* model’s own error differs up to fivefold between the two window sets), so this is a protocol-level failure, not a mechanism (App. G). The prescription is unsurprising; that ignoring it inverts 4 of 31 conclusions, by up to 46 pp and always in the same direction, is not.

The criterion says where the question is meaningful, not where fine-tuning will fail. The gate inherits selection the same way, and the eight cells below were identified *retrospectively*, so we **pre-registered** the criterion on eight new cells, committed to git before any run. It **flagged all eight; two degraded: precision 0.25 at recall 1.00**. The gate defines the population in which the counterfactual question is meaningful; our prospective experiment gives no evidence that it can identify degradation.

Finding 2: drift magnitude does not order the intervention. Sharpest is a near-tie: two Moirai-Large $h=96$ cells at CKA 0.622/0.626 sit at B–D $+12.0/-20.8$ pp, same backbone, size and horizon, 33 pp apart, and no cut separates them. Nor is there reliable evidence that CKA *orders* the held-out column *within* a backbone: $\rho = +0.168$ over 22 Moirai cells, on a CI wide enough to leave a moderate positive effect open, and the slope stays unpinned when backbone identity is absorbed as a fixed effect ($n=31$, App. F). This is evidence of **non-reliability** under the evaluated protocols, not proof of zero association.

Finding 3: fine-tuning can leave the model worse than not fine-tuning at all. In eight Moirai cells, full encoder fine-tuning *destroys* the pretrained advantage the criterion demonstrated, while freezing the encoder keeps it, so the practitioner would have been better off leaving the checkpoint alone. We name this regime **pretrained-capability degradation** and define it **empirically** by three clauses, the last two required in *every* seed: the cell gate-passes, B ends up worse than the un-tuned model, and D improves on it. Moirai-Small/Weather is clearest: B lands $+37.3\%/+46.1\%$ above zero-shot error while D reaches $-15.8\%/-16.0\%$. **On ETTh1 it holds at all three released sizes** (Large prospectively, $+14.1\%$ under B against -14.1% under D, 3/3); on Weather it is present at Small and again at Large but absent at Base, so neither dataset nor size predicts it alone. Nor does drift flag them: the eight span CKA 0.36–0.89 while the seven cells adaptation helps by ≥ 20 pp span

Table 1: **Does CKA tell us whether freezing helps? No: the CKA column does not order the B–D column.** Held-out B–D (pp): <0 **adaptation helps**, >0 **freezing helps**. 31 cells, 3–10 seeds, scored on test. ^fgate-fails; ^{*} $|\text{mean}| < 2 \text{ SEM}$; [‡]B–D only (limitation iii); ^ppre-registered. Per-cell gate values and B–D App. G; statistics App. F.

Backbone	Data (h)	CKA	B–D held-out	Reading
Chronos [‡]	ETTh1, ETTm2, Weather, ETTh2, Elec. ^f (24)	0.09–0.23	–39.2 to –2.7*	adapt
Moirai-S	ILI (24) ^f	0.455	–38.6	adapt
Moirai-B	ETTh2 (96 / 192)	0.40–0.46	–22.3 / –20.2	adapt
Moirai-L	ETTh2 (96) ^f	0.626	–20.8	adapt
TimesFM	ETTh2, ETTh2, Weather, ETTh1 (24)	0.25–0.56	–7.7 to –1.4*	adapt
Moirai-S ^p	Electricity (96 / 192)	0.94–0.95	–3.2 / –2.6*	adapt
Moirai-S	ETTh2, ETTm2 (96 / 192)	0.85–0.97	+2.6 to +6.1*	freeze
Moirai-B ^p	ETTh2, Weather (96 / 192)	0.49–0.57	+1.3* to +12.5*	freeze
Moirai-L ^p	Weather (96)	0.622	+12.0*	degradation
Moirai-S, L ^p	ETTh1 (96 / 192)	0.36–0.84	+13.5 to +28.2	degradation
Moirai-B	ETTh1 (96 / 192)	0.67–0.76	+37.5 / +47.0	degradation
Moirai-S	Weather (96 / 192)	0.88–0.89	+53.1 / +62.1	degradation

0.09–0.63, so the bands **interleave** and the reflex would have cleared damaging cells and flagged safe ones.

Robustness. Freezing the input projections too (condition H, CKA exactly 1.0) keeps B–D’s sign on all seven cells run (six of the eight degradation cells, plus ILI), so “freezing helps” is not input re-fitting (App. H). **A third backbone.** TimesFM-2.5 under its *own* recipe (no head attached, so both forgetting% terms are clean) gate-passes on four datasets and **none is a degradation cell**: B beats zero-shot in all four, B–D running –7.7 to –1.4 (scope, not generality). **Representation similarity is not a property of the backbone weights alone.** In condition D TimesFM’s transformer stack is **bit-identical** after training (ℓ_2 drift exactly 0.00) yet reads CKA 0.38–0.89, all of it the input tokenizer, so low CKA cannot by itself mean a damaged encoder (App. E).

4 Discussion

What this is and is not. The sign reversals are a generalization result: no model is fitted or selected on test. The degradation regime is *identified* using test data, so it establishes existence, not prevalence; the pre-registered arm bounds its detection accuracy (App. G). Runs were not blinded and two pre-specified predictions were falsified.

The one sentence to carry away. For TSFM adaptation, representation drift is an *observational diagnostic*, not a substitute for the counterfactual question of whether *allowing* encoder adaptation improves held-out utility.

Practical takeaway: do not infer damage from drift.

- 1. Check capability.** Does the released checkpoint beat a lookback-96 linear baseline?
- 2. Test adaptation.** Compare full fine-tuning against a frozen encoder.
- 3. Use held-out data.** Score B–D only on a chronologically disjoint test split.

Scripts, per-seed run records and the pre-registration (committed before any run) will be released with the paper.

Limitations (related work App. K). (i) Two analyses from an earlier version are **withdrawn**, sharing no machinery with B–D (App. B, F). (ii) Cells are unequally powered and **not independent**, so cross-cell statistics are descriptive. (iii) Chronos carries no per-condition forgetting% ($\ddagger$, Table 1); the degradation cells are Moirai-only. (iv) Condition D leaves input projections trainable, and the strict-freeze control (App. H) covers six of the eight cells: five keep both parts of the degradation definition, but on Moirai-Small/ETTh1 $h=192$ the strictly-frozen model no longer beats the un-tuned one.

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191 A Within-Model Dissociation Sweep

192 Every point shares a backbone, a dataset, a horizon and a protocol; only the fine-tuning sample size
 193 varies. Increasing n moves drift monotonically while the task outcome moves non-monotonically:
 194 CKA falls steadily from 0.949 at $n=500$ to 0.518 at $n=10k$, while forgetting goes +14.0, +13.4,
 195 −2.9, +6.6, −5.3%. The best cell in the sweep is the *most*-drifted one. The $n=5k$ variance (+6.6%,
 196 std 10.0, 7 seeds) is bimodal: 4 seeds near −0.1%, 3 at +15.5% with an epoch-1 overshoot, so we
 197 treat that point as unresolved rather than as evidence for either direction.

198 This is a *dissociation*, not an *intervention on drift*. Sample size is an uncontrolled causal variable:
 199 raising it changes supervision, optimization trajectory and overfitting, not drift alone. The sweep
 200 therefore shows that drift and outcome need not move together within one model; it cannot show that
 201 drift itself is causally inert. The frozen-encoder experiment is the intervention, and it is what the
 202 paper’s claims rest on.

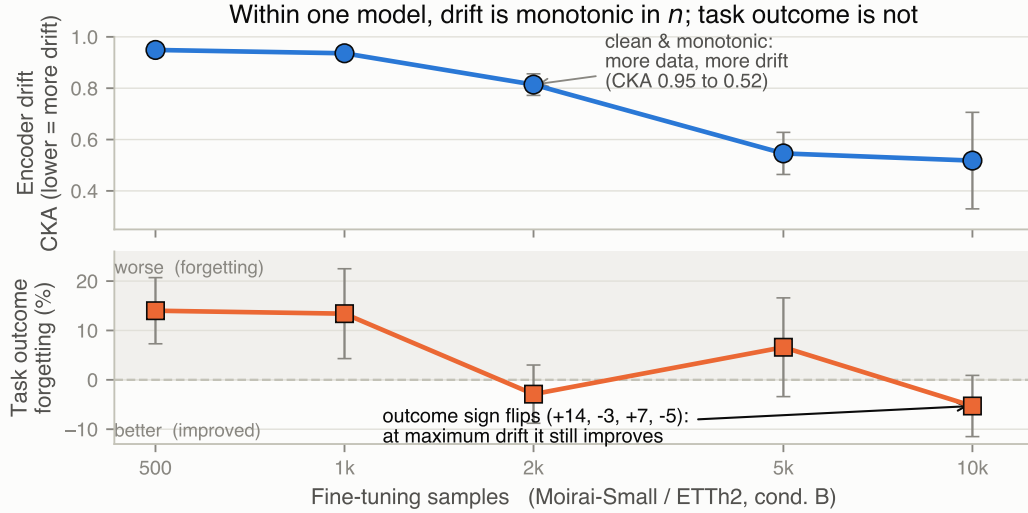


Figure 2: **Within-model dissociation.** Moirai-Small/ETTh2 ($h=96$, cond. B) across n : as n rises CKA drift is monotonic (top) while forgetting is non-monotonic and sign-flipping (bottom), improving at maximum drift: **drift and downstream outcome can move independently**. Sample size is uncontrolled here, so this demonstrates **non-monotonic co-movement, not causal independence**: it is a dissociation, not an intervention on drift. Error bars ± 1 SD; $n=10k$ is 10-seed CUDA-deterministic.

203 B The Retracted Probe Analysis

204 We initially used a linear-probe analysis. We reproduced the published statistic but found that it
 205 depended on a non-generalizing probe ($R^2 \approx -6.9$) and disappeared under a protocol in which the
 206 probe generalizes. We therefore **retract this analysis**. It shares no data-processing machinery with
 207 the frozen-encoder intervention, which manipulates the encoder rather than fitting a readout on it and
 208 is measured in forecast MSE rather than R^2 ; the reversals in §3 were never probe-dependent. The
 209 numbers are kept here, unchanged, so the retraction can be audited.

210 **What was published, and that it reproduces.** On Moirai-Small/ETTh2 ($h=96$, $n=10k$) the
 211 trained 96-step probe gave $\Delta R^2 = +0.67 \pm 0.23$ (10/10 seeds positive) while four auxiliary probes
 212 decreased: mean -0.033 ± 0.019 (10/10 neg), variance -0.048 ± 0.030 (9/10), lag-1 autocorrelation
 213 -0.052 ± 0.041 (9/10), GBM delta1 -0.056 ± 0.030 (4 seeds), and forecast-48 was indistinguishable
 214 from zero (-0.001 ± 0.163 , 4 seeds). Classification used an effect-size heuristic, whether the mean
 215 exceeds one seed-level SD, not a significance test. The five targets vary along two axes a single
 216 probe family would confound: *what* is decoded (low-order statistics vs. forecasts) and *how* (linear
 217 Ridge vs. gradient-boosted). The pipeline was not documented in enough detail to re-run, so we
 218 recovered it from source: encoder input of **192** steps (lookback + horizon), probe trained on the first
 219 300 validation windows and evaluated on the first 200 test windows, target the next 96 steps across
 220 all seven features flattened, Ridge($\alpha=1.0$) on mean-pooled outputs. That reproduces the published
 221 values to four significant figures: $R^2(\text{PT}) = -6.9043$ (published -6.90), $R^2(\text{FT}) = -6.2364$ (-6.23),
 222 $\Delta R^2 = +0.6679 \pm 0.2262$ ($+0.67 \pm 0.23$), 0/10 negative. Three details each move the result by an
 223 order of magnitude: the 192-step window rather than a 96-step context, evaluating across the
 224 validation $\rightarrow$ test boundary rather than within one split, and using all seven features rather than the
 225 target channel alone; a within-validation split moves $R^2(\text{PT})$ from -6.90 to -0.13 .

226 **A protocol with every knob fixed in advance** (`scripts/probe_transparent.py`): the same
 227 500 windows; mean-pooled encoder output standardized on probe-train; Ridge with α chosen by
 228 5-fold CV on the *pretrained* encoder’s probe-train over $\log_{10} \alpha \in [-3, 6]$ and then reused for all
 229 ten fine-tuned encoders, so the two are never tuned apart; ΔR^2 per seed with a percentile bootstrap
 230 95% CI over the ten seeds (10,000 resamples). Two additions decide the outcome: **two splits** (T , the
 231 published temporal split, and R , the same 500 windows randomized into 300/200), which separates
 232 decodability from the validation→test level shift, and a **permuted-target noise floor**, the identical
 233 pipeline run on 20 independently shuffled copies of each target, so a real ΔR^2 must exceed what the
 234 pipeline produces when the target carries no information at all.

Table 2: Transparent protocol, Moirai-Small/ETTh2, 10 released checkpoints. Under the published split T no probe generalizes ($R^2(\text{PT}) < 0$) and every $|\Delta R^2|$ is far below the permuted-target floor. Under R , where every probe generalizes, no $|\Delta R^2|$ exceeds 0.011 and the trained probe *decreases*.

Split	Target	$R^2(\text{PT})$	ΔR^2	95% CI	floor
T	Forecast-96 (trained)	−5.23	−0.0005	[−0.001, +0.000]	0.0005
	Mean	−164.09	+3.22	[−3.60, +10.00]	102.1
	Variance	−20.50	−8.22	[−11.94, −4.14]	132.3
	Lag-1 autocorr.	−2.71	−0.80	[−2.02, +0.16]	30.7
R	Forecast-96 (trained)	+0.678	−0.0042	[−0.0061, −0.0025]	0.019
	Mean	+0.994	−0.0014	[−0.0022, −0.0006]	0.226
	Variance	+0.937	+0.0025	[−0.0014, +0.0068]	0.169
	Lag-1 autocorr.	+0.750	+0.0103	[+0.0012, +0.0194]	0.223

235 **The effect is a regularization artifact.** Holding the split at T and sweeping α , the trained
 236 probe’s ΔR^2 decays monotonically: +0.668, +0.271, +0.066, +0.034, +0.005, +0.000 at
 237 $\alpha = 1, 10, 10^2, 10^3, 10^4, 10^5$, while $R^2(\text{PT})$ stays between −6.90 and −5.23. It is largest exactly
 238 where the probe extrapolates worst, and CV on the training split selects $\alpha=10^6$, at which it vanishes.
 239 Under R , where the trained probe attains $R^2(\text{PT}) = +0.68$, ΔR^2 is at most +0.017 at any α and
 240 is negative at the CV-selected value. No configuration has the trained probe both generalizing and
 241 increasing by anything near +0.67. Target normalization is provably not a knob: with an intercept,
 242 Ridge is equivariant under affine target scaling, and the sweep confirms the rows are identical to the
 243 last digit. The criterion that disqualified the Ridge delta1 probe was that its zero-shot baseline is
 244 negative enough ($R^2 \approx -2.3$) that any encoder shift inflates ΔR^2 , which is why we reported the
 245 GBM version on a genuine positive floor ($R^2(\text{PT}) = +0.059$, on which decodability *falls*) rather than
 246 Ridge’s +1.04±0.22. Applied consistently, that criterion disqualifies the headline trained probe at
 247 −6.90 as well.

248 **What survives.** Under the one split where all seven probes generalize, every $|\Delta R^2|$ is ≤ 0.011 :
 249 the trained probe *decreases* (−0.0042, 10/10 seeds), mean and trend decrease, lag-1 autocorrelation
 250 *increases*, and variance, lag-24 and spectral centroid are indistinguishable from zero. That is not
 251 the claimed asymmetry; it is closer to no detectable change in any direction. Five probes could not
 252 have established the absence of specialization along an unmeasured axis (calibration, periodicity,
 253 tail behaviour) in any case, and the GBM delta1 probe was the only positive-floor untrained readout
 254 available to us.

255 C ILI Cell Detail

256 **ILI gate-fails on held-out windows.** Computed from the runs this paper reports (10 seeds,
 257 `results/v40_ili_heldout`), the gate is +0.139 on validation (ZS 1.893, linear 2.206) and
 258 **−0.033** on the test segment (ZS 2.479, linear 2.664); an earlier gate-pass figure is withdrawn.
 259 The per-feature picture is why the mean lands where it does: five of seven features are positive but
 260 only two clear 0.20, and NUM. OF PROVIDERS is −0.86 because that series is close to a straight line
 261 on the test segment, which linear extrapolation fits almost exactly. ILI is not a degradation cell and
 262 no claim in §3 rests on its gate; what changes is that it can no longer be cited as a moderate-gate
 263 cell, and the “falsified prediction” below is correspondingly weaker evidence: a heavily restructuring
 264 encoder on a cell with *no* demonstrated zero-shot advantage is much less surprising than one on a
 265 moderate-gate cell.

Status of this cell. On ILI the encoder restructures heavily (CKA 0.455 ± 0.008) and full fine-tuning improves by $-80.0 \pm 0.6\%$ (10/10), falsifying a prediction specified beforehand (0.90–0.95 CKA expected). We previously reported B–D = -80.7 pp here, i.e. that encoder adaptation supplied *all* of the gain. That is withdrawn. The frozen control it was measured against had never been trained: the script assigned the zero-shot MSE as the fine-tuned MSE and wrote a forgetting of exactly 0.0, so the “0.0% in all 10 seeds” was one constant, not ten measurements. Re-run so that condition D actually trains (encoder frozen, the 3.2M parameters outside it updated), the gap is -62.6 ± 1.4 on validation and -38.6 ± 1.3 held-out (10/10 negative on both). D reaches CKA 0.76–0.90, not 1.000, because the projections feeding the encoder remain trainable. Condition B is unaffected and reproduces the published -80.7 ± 0.9 and CKA 0.478 to within seed noise, so the entire discrepancy was the untrained control. Note also that both references are weak in absolute terms (ZS MSE 1.893, linear 2.206 on validation).

The cell is Moirai-Small on national influenza-like illness: 966 weekly observations, 7 features, $h=24$, lookback 104, split 579/193/194 chronologically. Condition D, re-run so that it trains, gives CKA 0.827 ± 0.013 and forgetting $-17.4 \pm 1.3\%$ on validation ($-12.2 \pm 1.3\%$ held-out), 10/10 negative: head-only training recovers a substantial share of the improvement, so the B–D gap is -62.6 pp on validation and -38.6 pp held-out, not the -80.7 previously reported.

Table 3: ILI per-seed results (Moirai-Small, $h=24$, lookback 104), both conditions, regenerated for this revision. Each value is the mean across the 7 univariate features. The ΔR^2 column of the earlier version of this table is dropped with the retracted probe axis; condition D is shown because it previously did not exist as a measurement.

Seed	B (full fine-tuning)		D (frozen encoder)	
	CKA	Forg.% (val / held-out)	CKA	Forg.% (val / held-out)
42	0.458	$-78.6 / -51.5$	0.804	$-16.2 / -14.6$
101	0.465	$-81.4 / -50.1$	0.898	$-13.5 / -6.7$
123	0.476	$-80.0 / -48.9$	0.756	$-27.6 / -19.5$
202	0.457	$-81.4 / -52.1$	0.791	$-19.7 / -13.6$
303	0.409	$-78.1 / -49.0$	0.823	$-18.3 / -11.5$
456	0.463	$-75.5 / -50.0$	0.851	$-15.4 / -13.2$
777	0.437	$-81.6 / -52.5$	0.822	$-18.0 / -12.4$
789	0.449	$-80.3 / -49.5$	0.865	$-12.7 / -4.9$
888	0.505	$-82.0 / -50.8$	0.848	$-16.1 / -12.8$
999	0.437	$-80.8 / -53.4$	0.807	$-16.1 / -12.7$
Mean	0.456	$-80.0 / -50.8$	0.827	$-17.4 / -12.2$

Per-feature asymmetry. The forecast target OT shows the largest gain ($+0.73 \pm 0.16$, 10/10 positive) and AGE 0–4 also gains consistently ($+0.15 \pm 0.05$, 10/10). AGE 5–24 is negative (-0.07 ± 0.09 , 3/10 positive). Decodability thus improves on the target and degrades on a series the head was not optimizing, the same signature as the ETTh2 trained-vs-auxiliary split, obtained here within a single cell.

Falsified prediction. We specified CKA 0.90–0.95 for ILI in advance, on the expectation that a moderate gate implies moderate drift. The observed 0.456 falsifies this: gate strength does not linearly predict drift magnitude, and the restructuring-to-stability transition sits nearer $\approx 80\%$ gate than the 50–70% we assumed. We report this because it is the kind of prediction the chain is supposed to expose.

D Full Preservation Spectrum

What this appendix is for. It is a *secondary*, observational result: it shows that raising representational similarity does not raise utility, which is evidence against the reflex’s prescriptive half. It is **not** evidence for this paper’s interventional claim; that rests entirely on the frozen-encoder intervention B–D in §3, because these methods preserve representations by different mechanisms and so are not a controlled manipulation of one variable. The main text quotes $h=96$ only. Table 4 gives both horizons and all methods, including the NLL+SupCon variant. These methods preserve representations by *different mechanisms*, namely penalty (L2-SP, EWC), reparameterization (LoRA), and complete

Table 4: Mitigation spectrum on ETTh2 forecasting. Methods ordered by representation preservation (CKA, $h=96$). LoRA preserves representations (CKA ≈ 0.98) while *improving* over zero-shot. Mean $\pm$ std; Forgetting% = (fine-tuned MSE – zero-shot MSE) / zero-shot MSE $\times$ 100. **Two caveats on comparability, added in revision.** Forgetting% divides by the single shared zero-shot printed in each horizon header, not by each run’s own zero-shot; the anchoring and LoRA runs measure their own at 0.127–0.131, so their figures here are inflated by 1.2–3.4 pp. And the rows are not matched on training-set size: B and C use $n=500$ training samples while L2-SP, EWC and LoRA use $n=1000$ (the n column counts seeds, not samples). Matched at $n=1000$, $h=96$, each against its own zero-shot: full fine-tuning .936/+13.4, L2-SP $\lambda=0.01$.939/+16.0, L2-SP $\lambda=0.1$.965/+13.4, EWC .972/+14.9, LoRA .979/−9.2. On that comparison L2-SP $\lambda=0.1$ ties full fine-tuning rather than losing to it, so anchoring is not uniformly harmful; the dissociation the table is cited for (CKA not predicting utility) is unaffected.

Method	$h = 96$ (ZS MSE: 0.126)				$h = 192$ (ZS MSE: 0.158)				n
	MSE	Forg.%	CKA	Drift	MSE	Forg.%	CKA	Drift	
B: Full fine-tune	.140 $\pm$.014	+11.1	.937 $\pm$.028	3.96 $\pm$.56	.183 $\pm$.020	+15.7	.970 $\pm$.009	4.12 $\pm$.32	10
F: L2-SP ($\lambda=0.01$)	.150 $\pm$.006	+19.0	.939 $\pm$.023	4.98 $\pm$.34	.179 $\pm$.009	+13.3	.945 $\pm$.007	4.84 $\pm$.22	5
C: NLL+SupCon	.138 $\pm$.010	+10.1	.949 $\pm$.014	3.72 $\pm$.43	.178 $\pm$.011	+12.9	.964 $\pm$.015	3.97 $\pm$.34	10
F: L2-SP ($\lambda=0.1$)	.147 $\pm$.003	+16.7	.965 $\pm$.006	2.94 $\pm$.08	.181 $\pm$.003	+14.6	.980 $\pm$.002	2.84 $\pm$.02	5
G: EWC ($\lambda=1000$)	.149 $\pm$.015	+18.3	.972 $\pm$.004	3.29 $\pm$.79	.188 $\pm$.004	+19.0	.978 $\pm$.006	4.46 $\pm$.38	5
E: LoRA ($r=8$) [†]	.116 $\pm$.001	−7.9	.979 $\pm$.006	0.00 $\pm$.00	.157 $\pm$.002	−0.6	.986 $\pm$.005	0.00 $\pm$.00	5
D: Frozen encoder	.126 $\pm$.006	+0.5	1.000 $\pm$.000	1.11 $\pm$.15	.155 $\pm$.002	−1.7	1.000 $\pm$.000	1.30 $\pm$.18	10

[†] LoRA ($\alpha=16$) adds low-rank updates that do not modify base encoder parameters; ℓ_2 drift is exactly zero by construction.

freezing, so CKA is a common observational axis, not a common intervention, and at $n=4$ –7 the correlations here are descriptive only. We stress that **LoRA does well**: the paper’s claim is that drift magnitude fails to *order* these methods, not that preservation is a bad idea or that fine-tuning should be avoided. **The matched spectrum** puts every method at $n=1000$, $h=96$, each run against its own zero-shot: full fine-tuning is CKA .936 at +13.4%, against L2-SP $\lambda=0.01$ (.939, +16.0), L2-SP $\lambda=0.1$ (.965, +13.4), EWC $\lambda=1000$ (.972, +14.9) and LoRA (.979, −9.2).

The claim this displaces is the strong one: it is *not* true that every weight-anchoring configuration forecasts worse. L2-SP $\lambda=0.1$ raises CKA substantially (.936 $\rightarrow$.965) and lands on full fine-tuning to within 0.02 pp. What the corrected spectrum shows is the weaker and more useful thing, that moving up the CKA axis buys nothing predictable: one anchoring setting is worse, one is a dead heat, a third is worse again, and the reparameterizing method is far better while being the most CKA-preserving of the four. Representation preservation does not predict utility here. At $h=192$ no $n=1000$ full-fine-tuning baseline exists in our runs, so we report that horizon’s methods without a matched reference rather than against the $n=500$ one.

Why we do not report a single CKA–utility correlation. A natural request is a correlation over Table 4 rather than a verbal reading of seven points. We computed it, and the answer is instructive: *the sign depends on which mechanisms are pooled.*

Conditions included	$h=96$		$h=192$	
	Pearson r	Spearman ρ	Pearson r	Spearman ρ
Weight-anchoring only ($n=4$)	+0.46	+0.40	+0.64	+0.40
All seven conditions ($n=7$)	−0.58	−0.54	−0.60	−0.43

Forgetting is negative-is-better, so a *positive* coefficient means higher CKA accompanies worse utility. Among weight-anchoring methods the sign is positive at both horizons; pooling all seven flips it negative, because LoRA and the frozen encoder sit at the highest CKA and perform well. Neither coefficient is meaningful on its own at $n=4$ or $n=7$, and we claim no significance. The point is the flip: CKA is a common *observational* axis across these methods, not a common intervention, so no single correlation summarizes the table. This is the same reason we present the spectrum as a mechanism-separated comparison rather than a regression.

E Chronos and TimesFM Intervention Detail

Per-arm objectives and heads. The 22 Moirai cells (which include all eight degradation cells) plus ILI train through Moirai’s own NLL objective and distribution head (`param_proj`), scored on its native sampling path; the 4 TimesFM cells likewise use TimesFM’s own point/quantile head. No regression head is attached in either arm. Only the 5 Chronos cells attach a common MSE head to encoder states, because its generative decoder admits no matched frozen-versus-adapted comparison otherwise; hence limitation (iii). What departs from the released recipes is the *data* pipeline, not the objective, head or inference path; the cost is that MSE magnitudes are *not* comparable across arms, which is why B–D is read within a backbone.

Electricity restriction. *Electricity is restricted to 7 of its 370 series* so its feature dimensionality, and hence the cost of a multivariate Moirai run, matches the ETT cells. This is a deliberate deviation from the standard benchmark, and its numbers are therefore *not* comparable to published full-Electricity results.

Chronos-T5-Small, moderate gate ($n=8k$, conditions B and D). This pair is the main text’s first inversion. Both conditions are trained to their *own* convergence rather than to a shared 30-epoch budget. That budget is generous for B, which early-stops with its best epoch at 1–2, but it truncates D badly: under it every D seed restored “best epoch 30 of 30” with validation loss still falling nearly linearly, so the gap it measured was the budget rather than the intervention.

Because in condition D the encoder is frozen *and* in `eval()` mode, its mean-pooled output is a deterministic function of the context window, and the head is a single linear map on that vector. Caching the pooled features is therefore exact, not an approximation, and it makes convergence affordable. We then estimate the best head two independent ways: a closed-form ridge optimum over 16 log-spaced penalties (the headline, since a closed form admits no epoch budget at all; the penalty is selected on the same eval windows B early-stops on, so both sides receive eval-set selection, and the optimum was interior to the grid in all six runs), and a protocol-matched AdamW run to early stopping. The two agree to about 1 pp everywhere, which is the evidence that the number is a property of the frozen features rather than of the optimizer.

Table 5: Chronos-T5-Small, $h=24$, $n=8k$, matched seeds and eval windows. MSE is per-element validation MSE; $B-D=(MSE_B - MSE_D)/MSE_{ZS} \times 100$, so the zero-shot reference cancels and only the two same-head models are compared. All six D runs converged (early stopping fired) and all six ridge optima were interior.

Cell	Seed	MSE _B	D (ridge optimum)		B–D (pp)		
			MSE _D	α	ridge	AdamW	CKA (B)
ETTh1	42	1.598	1.580	0.3	+1.2	+0.5	0.179
	43	1.471	1.364	0.03	+7.5	+5.9	0.158
	44	1.509	1.365	0.1	+11.4	+11.1	0.171
mean±std					+6.7±5.1	+5.8±5.3	0.169
ETTh2	42	0.998	1.133	0.03	−13.3	−14.8	0.094
	43	0.842	1.053	0.03	−23.4	−25.0	0.102
	44	0.882	1.078	0.03	−22.8	−21.7	0.087
mean±std					−19.9±5.7	−20.5±5.2	0.094

One further consequence of the re-run is why Table 1 reports B–D for these cells but no per-condition forgetting. Forgetting is a ratio against the zero-shot reference, and for Chronos the fine-tuned model is scored through the trained MSE head while that reference is scored through Chronos’s generative decoder, so the ratio charges the output-head swap to forgetting and is not a measurement of adaptation at all. B–D is immune because both conditions share the head and the reference cancels algebraically. We therefore report no per-condition forgetting for Chronos. The measured condition B values are +12.6% on ETTh1 and −1.6% on ETTh2, and the measured Chronos CKA range is 0.087–0.179.

On the validation windows every ETTh1 seed sits above every ETTh2 seed, but that separation does not survive held-out scoring: on the test windows both cells are negative (ETTh1 −39.1±24.1, ETTh2 −12.7±1.0, 3/3 each), so we report no sign contrast between them (App. G). What the two

cells still show is that CKA does not order the outcome: ETTh1 drifts *less* (0.158–0.179 against 0.087–0.102) and loses *more* to freezing.

Chronos-T5-Small, strong gate. On M4-Monthly (gate 84.5%, 10 seeds) condition B gives +0.5% and condition D +0.6% (10 seeds; std 2.7 and 5.3); the B–D gap is within the frozen-control noise floor, and a sign test on the 6/10 positive-forgetting seeds is not distinguishable from chance ($p=0.75$, two-sided binomial).

Cross-dataset transfer. Fine-tuning Moirai-Small on ETTh2 improves ETTh1 ($-7.7 \pm 2.5\%$) and ETTm2 ($-3.3 \pm 1.0\%$), 3 seeds, condition B; the frozen control shows near-zero out-of-distribution effect.

Pretraining overlap. Moirai’s LOTSA corpus excludes ETT; M4 is in Chronos’s pretraining data (which is why the M4 cell functions as an overlap control); overlap for TimesFM and MOMENT could not be established from public documentation.

TimesFM-2.5-200M at $h=96$. Five seeds on ETTh1 across conditions B and D give stability throughout (CKA 0.995–0.998, $\Delta R^2 \approx 0$, no asymmetry), and the checkpoint gate *fails* at this horizon ($R^2_{\text{task}} -0.139$ on ETTh1, -0.138 on ETTh2), so the inclusion criterion does not license an intervention there. Per-seed forgetting for the two conditions was not retained, so the pair sits outside the 31 intervention cells of Table 1 rather than being entered with an estimate.

TimesFM-2.5-200M at $h=24$: the criterion. Because the criterion is horizon-sensitive, we re-screened TimesFM at $h=24$ before running any intervention, on the *same* held-out window set the Chronos arm uses (univariate OT, lookback 96, the 200-window seed-0 test subsample), so the criterion, the zero-shot reference and B–D all read one window set. All five datasets pass, and they track Chronos dataset for dataset, which is evidence that R^2_{task} is measuring a property of the dataset $\times$ horizon rather than of the backbone:

	R^2_{task} at $h=24$ (test)	ETTm2	Weather	ETTh2	ETTh1	Electricity
387	TimesFM-2.5-200M	+0.703	+0.506	+0.472	+0.401	+0.214
	Chronos-T5-Small	+0.651	+0.494	+0.456	+0.389	+0.138

Both arms are scored by `scripts/gate_all_cells.py`, one code path for all four arms, cached to `results/gate_test_side.json`.

TimesFM-2.5-200M at $h=24$: the intervention. Having licensed it, we run the same conditions as the Moirai arm (`scripts/finetune_timesfm.py`): **B** trains everything; **D** freezes the 20-layer stack (196.71M of 231.29M parameters) and leaves the input tokenizer and output projections trainable, exactly as Moirai’s D leaves `in_proj` trainable. Hyperparameters are the ones the other arms use (lr 10^{-4} , AdamW, grad-norm clip 1.0, 20 epochs, patience 7, 1000 training windows) and are identical for B and D, since B–D would be meaningless otherwise. After restoring the best epoch we *assert* the frozen modules are bit-identical to the released checkpoint; the run aborts rather than mislabel a condition.

Two design points matter for reading these cells. First, no head is attached: the loss is TimesFM’s own output structure (MSE on the mean channel plus pinball loss on the nine decile channels the released `output_projection_point` emits), and zero-shot, B and D are all scored through the same native path. **Per-condition forgetting is therefore a clean measurement here**, unlike Chronos (limitation iii), so this arm re-tests the `forB` rule on a second backbone. Second, the released torch API cannot be trained through: `module.decode()` is wrapped in `torch.no_grad()` and returns detached NumPy, so training uses a differentiable re-implementation of the prefill path. That path is legitimate only because at $h=24$ the autoregressive loop runs $(\text{max_horizon} - 1) // o = 0$ times with $o=128$: one forward pass *is* the whole forecast. `verify_native_path()` asserts agreement with `model.forecast()` on real windows before any training (max relative deviation 6.5×10^{-7} on MPS, 0.0 on CPU), so a re-implementation error cannot reach the paper silently. The continuous quantile head (27.85M) rewrites only non-median channels at inference and receives no gradient, so condition B updates 203.44M of the 231.29M parameters.

What the four cells show. We ran the four highest-gate datasets, 3 seeds $\times$ {B, D} each, in an order fixed before any run by correspondence to the degradation cells (ETTh1 first, then Weather, then ETTm2, ETTh2). **All four land in the *adapt* half of Table 1, and none is a degradation cell:** condition B improves on zero-shot everywhere (`forB` -51.8% to -3.0%), which is the first clause

of degradation failing, so the regime does not extend to this backbone at $h=24$. That is a scope limit on H3 (limitation iv), not a refutation of it: the degradation cells sit at $h=96/192$, where TimesFM gate-*fails* and the inclusion criterion forbids the comparison.

	TimesFM, $h=24$	gate	CKA (B)	CKA (D)	for _B	for _D	B–D val	B–D held-out
ETM2		+0.703	0.19–0.52	0.48–0.55	-51.8 ± 2.8	-44.1 ± 1.6	-12.2 ± 12.4	-7.7 ± 2.6
Weather		+0.506	0.24–0.40	0.62–0.71	-7.8 ± 1.6	-6.1 ± 0.5	-4.8 ± 0.6	-1.6 ± 1.5
ETTh2		+0.472	0.22–0.28	0.38–0.50	-20.4 ± 1.4	-13.3 ± 1.9	$+0.2 \pm 2.9$	-7.1 ± 3.1
ETTh1		+0.401	0.44–0.64	0.80–0.89	-3.0 ± 2.5	-1.6 ± 0.9	-5.4 ± 0.9	-1.4 ± 2.8

Three things the arm does establish, none of which was available from Moirai or Chronos.

(a) The held-out reversal replicates on fresh cells. These four cells were scored on both splits from the same runs (nothing was re-scored after the fact), and TimesFM/ETTh2 *reverses sign*, $+0.2$ on validation against -7.1 held out. It reverses in the *same* direction as the three Moirai/Chronos reversals: validation favoured the frozen encoder, held-out data does not. Four of our 31 cells reverse and all four reverse that way, which is the paper’s central methodological claim (§3) reproduced on a backbone it was not derived from.

(b) A stack that is provably unchanged can read CKA 0.38. Condition D’s ℓ_2 weight drift over `stacked_xf` is exactly 0.00 (the bit-identity assertion above), yet its measured CKA runs 0.38–0.89. Every point of that movement comes from the input tokenizer changing what the frozen stack is shown. This is the sharpest form of the paper’s measurement claim: CKA registers representational change on a network whose weights did not move at all, so a low CKA cannot by itself be read as the encoder having been damaged. It also bounds limitation (v) from the other side: D is not representation-frozen anywhere, and here we can say exactly why.

(c) Drift does not order the four. CKA (B) spans 0.19–0.64, comparable to the degradation cells’ drift, while held-out B–D stays within 8 pp of zero; the ranking is not monotone ($\rho = +0.400$, $n=4$, CI $[-0.778, +1.000]$), and the *most*-drifted cell (ETTh2, CKA 0.22–0.28) is among the two where adaptation helps most. Because no head is attached, `forB` is a clean measurement here, so this arm also re-tests the App. J rule: its *sign* transfers (`forB` < 0 and B–D < 0 in 4 of 4 cells), but its stated mechanism does not, since `forD` varies here almost as much as `forB` (SD 19.2 against 22.0, versus 8.1 against 26.2 on Moirai). The rule is an empirical property of a nearly-constant frozen arm, and that property is Moirai’s, not a general one.

We do not re-screen MOMENT: its $h=96$ margin is -147.8% , and its released checkpoint has no pretrained forecasting head, so the zero-shot term the criterion needs does not exist without fitting a probe, which would make the “zero-shot advantage” an artifact of the probe. MOMENT therefore supports no claim in this paper.

F Cross-Cell Statistics, and a Pooled Analysis We Do Not License

Across the 31 cells of Table 1 we ask whether either observational diagnostic orders the held-out intervention. Spearman ρ with a 10,000-sample bootstrap CI over cells:

	Grouping	n	CKA range	ρ	95% CI
	Pooled across backbones	31	0.09–0.97	+0.567	$[+0.265, +0.759]$
	Moirai only	22	0.36–0.97	+0.168	$[-0.325, +0.595]$
	Chronos only	5	0.09–0.23	+0.100	$[-1.000, +1.000]$
	TimesFM only	4	0.25–0.56	+0.400	$[-0.778, +1.000]$
	ℓ_2 weight drift (pooled)	25	n/a	+0.118	$[-0.304, +0.522]$

The ℓ_2 row covers 25 cells rather than 31 because the five Chronos runs and ILI did not retain a weight-drift measurement. The correctly pooled value is the $+0.118$ tabulated above, and the Moirai-only figure is $n=13$, $\rho = +0.044$, CI $[-0.545, +0.651]$; both include zero.

Absorbing the backbone instead of subsetting it. The within-backbone rule can be read as convenient: we adopt the split that removes the significant relationship. So we also fit the pooled

cells directly, with backbone identity absorbed as a fixed effect rather than by subsetting, and CKA entering as a continuous regressor (`scripts/cka_fixed_effects.py`):

	Model for held-out B–D	n	b_1 (pp per unit CKA)	95% CI
456	$b_0 + b_1$ CKA (no backbone term)	31	+47.7	n/a
	$b_0 + b_1$ CKA + backbone FE	31	+39.1	[−14.4, +98.5]
	+ CKA $\times$ backbone	31	+39.9	[−17.9, +104.9]

CIIs are a 20,000-draw bootstrap resampling *clusters*, a cluster being a (backbone, dataset) pair, because datasets and checkpoints recur across horizons and sizes and an unclustered interval would be far too narrow; draws that leave a backbone unrepresented are discarded and the retained count is printed by the script. **Both adjusted intervals include zero.** The pooled rank correlation therefore does not survive absorbing backbone identity, and it is not the within-backbone restriction that removes it: the same conclusion follows from a model fitted on all 31 cells at once. What the data cannot do is pin the within-backbone slope down; that is weaker than, and must not be read as, a claim that CKA carries no information.

The pooled row is not a licensed analysis, and it points the other way. Pooled, CKA orders the intervention with a CI excluding zero ($p=0.0001$). We do not read it as evidence, for the reason stated in §2: CKA magnitude is representation-dependent, so values from different architectures are not on a common scale, and we compare them only within a backbone. That restriction was fixed before these cells were run, not chosen after seeing this table. The pooled figure is largely what an architecture confound would produce here: every Chronos cell sits at CKA 0.09–0.23 and every Moirai cell at 0.40–0.97, so pooling those two recovers the backbone split rather than a drift–utility relationship. The third backbone complicates that reading: TimesFM’s four cells span 0.25–0.56 and therefore *straddle* the Chronos and Moirai ranges instead of forming a third stratum, so the pooled correlation is not purely a between-backbone artifact. It is also the group where drift is largest relative to outcome (four cells at CKA 0.25–0.56 whose B–D all sit within 8 pp of zero), so adding it pushes the pooled ρ up while adding cells that individually contradict the reflex. Neither observation licenses reading the pooled row as evidence; both are why we print it.

We report it because a reader can compute it in a minute, and because the central claim survives on a protocol restriction rather than on the absence of any correlation in the numbers. Within Moirai, the only group with enough cells and enough CKA range to test, $\rho = +0.168$ with a CI that includes zero.

We state the bound explicitly, because it is easy to over-read. At $n=22$ the interval [−0.325, +0.595] does not exclude a *moderate positive* relationship. So this is not evidence that CKA carries no ordering information; it is an absence of reliable evidence that it does, and that is the only claim we make from it. **The paper’s claim does not rest on this interval.** It rests on a case that needs no interval at all: two Moirai-*Large* $h=96$ cells at CKA 0.622 and 0.626 (same backbone, same size, same horizon) sit at held-out B–D +12.0 and −20.8 pp. Whatever the population correlation turns out to be, no threshold on CKA separates those two cells, and separating cells is what a diagnostic would have to do to be used as one.

Within TimesFM, the one other backbone with a clean forgetting measurement, ρ is +0.400 on 4 cells with a CI spanning almost the whole range, too few cells to test anything, and reported so that the arm’s failure to help is on the record rather than omitted. The ℓ_2 drift result is cleaner: it shows no ordering even pooled ($\rho = +0.118$, CI [−0.304, +0.522]), which is the more robust form of the claim that observational summaries of fine-tuning do not substitute for running the intervention.

G Selection-Window Bias: Held-Out Re-Scoring of the Intervention

What selection-window bias is. Scored the natural way, B–D is read on the evaluation windows that model selection already used: condition B early-stops on them, and for the converged Chronos runs the frozen head’s ridge penalty is chosen on them. That makes B–D a controlled within-cell contrast but not an estimate of generalization. Each training script in this work already carves a chronologically disjoint test split and then discards it; here we score it.

The bias has a *direction*, which is why it is worth naming rather than treating as noise: every cell that reverses reverses the same way, from “freezing better” on the selection windows to “adaptation better” out of sample, so those windows flatter the frozen arm. **We do not claim a mechanism**, and this

design cannot supply one. Two explanations predict exactly this direction and we cannot tell them apart: *selection overfitting* (the frozen arm’s ridge penalty is chosen directly on those windows, while condition B’s early stopping selects along a training trajectory rather than over a hyperparameter) and a *validation-to-test shift* that happens to cost the frozen arm more. What follows bounds the question rather than settling it.

Decomposing the reversals per arm. Normalizing each arm’s error by the zero-shot error on *its own* split isolates that arm’s val→test movement, and the two movements close the gap identically: $(B-D)_{\text{test}} - (B-D)_{\text{val}} = \Delta_B - \Delta_D$. Both the identity and every value below are asserted against the published rows by `scripts/heldout_decomposition.py`. Thirty of the 31 cells decompose; ILI is out because its script stores aggregate percentages rather than per-split MSEs.

Cell	B–D val	B–D held-out	Δ adapted (B)	Δ frozen (D)	$z_{\text{test}}/z_{\text{val}}$
Chronos / ETTh1 ($h=24$)	+6.8	−39.2	+8.0	+53.9	0.98
Moirai-B / ETTh2 ($h=96, n=1000$)	+3.8	−22.3	+12.2	+38.3	2.31
Moirai-L / ETTh2 ($h=96, n=500$)	+7.2	−20.8	−52.9	−24.8	5.07
TimesFM / ETTh2 ($h=24$)	+0.2	−7.1	−11.7	−4.3	1.09

Two readings, the first of which is a warning. $\Delta_D > \Delta_B$ holds in all four cells *by construction* (a positive-to-negative reversal is $\Delta_B < \Delta_D$), so that ordering is not evidence of anything. What is not entailed is the *sign*. On Chronos/ETTh1 and Moirai-B/ETTh2 both arms end up worse than their own zero-shot and the frozen arm collapses ($\Delta_D = +53.9, +38.3$); on Moirai-L/ETTh2 and TimesFM/ETTh2 both arms end up *better* and the adapted arm merely gains more. In half the reversals, then, nothing in the frozen arm failed, which is a different phenomenon wearing the same sign change.

Second, the two window sets are not exchangeable. The *untrained* checkpoint (no fitting, no selection, no hyperparameter) has a test/validation error ratio spanning 0.52–5.07 across the 30 cells. Where an untrained model’s own error moves several-fold between two window sets, a four-cell contrast cannot separate a selection bias from a shift, and we do not try. (That ratio bounds exchangeability rather than measuring a shift: the sets differ in size and construction, and for the older Moirai cells the test reference is a dataset-level average while the validation one is per-run.) This is why the body reports a protocol-level failure and stops there.

Two kinds of evidence, and only one of them is prospective. These are separate claims and the paper does not conflate them. (A) *The protocol finding is genuinely out of sample*. No model is fitted or selected on the test split, so the sign reversals in §3 are a generalization result. (B) *The degradation regime is a retrospective discovery*. Its definition includes a gate that we *recompute* on the test split, so the eight cells are identified using data we have already looked at. That establishes existence; it is not a prospective validation of the criterion, and it says nothing about how often the regime occurs or how well the criterion would detect it in advance.

A pre-registered prospective test. Because (B) is the weaker claim, we measure it rather than caveat it. We selected **eight new cells** (two new model sizes, a new dataset, and horizons already in the matrix) that played no part in defining the regime, computed each cell’s gate from *validation* windows alone, and **committed the predictions to git before any condition B or D run existed**. The guarantee is structural, not procedural: the outcome needs forg_B and forg_D from runs that did not yet exist, and every quantity behind a prediction comes from train/validation only. Two rules were registered:

- **gate rule:** $R_{\text{task}}^2 \geq 0.20$ on validation $\Rightarrow$ at risk. The paper’s own criterion, pre-specified in §2.
- **dataset rule:** dataset $\in \{\text{ETTh1, Weather}\} \Rightarrow$ degradation. This is *post hoc*: among the published Moirai cells, dataset separates the regime perfectly while gate values overlap. We registered it as a *competitor*, not as a contribution, because a rule we noticed after the fact outperforming the one we published is itself evidence about the criterion. The two disagree on four of the eight cells, which is what makes the test informative rather than confirmatory.

Cell (new)	gate _{val}	for _B	B–D held-out	Degradation?	Predicted
base Weather ($h=96$)	+0.41	−11.7 (0/3)	+9.4±2.0	no	gate ×, data ×
base Weather ($h=192$)	+0.59	−18.1 (0/3)	+12.5±8.7	no	gate ×, data ×
base ETTh2 ($h=96$)	+0.30	−13.1 (0/3)	+7.0±2.4	no	gate ×, data ✓
base ETTh2 ($h=192$)	+0.26	−8.4 (0/3)	+1.3±3.4	no	gate ×, data ✓
small Electricity7 ($h=96$)	+0.71	−16.0 (0/3)	−3.2±1.5	no	gate ×, data ✓
small Electricity7 ($h=192$)	+0.76	−18.6 (0/3)	−2.6±2.1	no	gate ×, data ✓
large ETTh1 ($h=96$)	+0.35	+14.1 (3/3)	+28.2±4.9	yes	gate ✓, data ✓
large Weather ($h=96$)	+0.47	+7.9 (3/3)	+12.0±7.5	yes	gate ✓, data ✓

The criterion is a base-rate predictor. It flagged all eight cells; two degraded. That is **precision** 0.25 **at recall** 1.00: no false negatives, six false positives. The dataset rule scores 6/8, better but not a rule either: it is correct wherever it predicted *no* degradation and wrong on both Moirai-Base/Weather cells. Taken with the published cells, degradation appears on ETTh1 at Small, Base *and* Large, but on Weather at Small and Large while *skipping* Base, not even monotone in model size, so **we do not offer a prospective rule**. This is the clearest support we have for §2’s “inclusion, not prescription”: the criterion says where an intervention is *interpretable*, not where fine-tuning will hurt.

Two further observations. The highest validation gates in the entire matrix (0.712, 0.764, Moirai-Small/Electricity7) belong to cells where adaptation *helps* or the comparison is directional: if the gate carried risk signal, those are where it should have appeared. And for_B is negative in six of the eight new cells with *zero* positive seeds anywhere, so the regime is rarer than eight-of-31 makes it sound.

A correction to the gate. In the Moirai arm of `gate_all_cells.py` the linear denominator was built with an extended lookback of 2×96 for every cell, while `finetune_forecasting.py` evaluates with lookback + h . The two agree at $h=96$ and diverge at $h=192$, where the numerator and denominator of R^2_{task} were being averaged over different target windows. Corrected, six $h=192$ gate values move (largest, Moirai-Base/ETTh2 0.72 $\rightarrow$ 0.55; Moirai-Small/Weather 0.80 $\rightarrow$ 0.83). **No cell changes gate status and the degradation set is unchanged**; the values in Table 1, Table 6 and App. M are the corrected ones.

Protocol. Train on train, select on validation, report on test. Nothing is selected on the test windows: condition B’s early stopping is untouched, and the ridge penalty is still the argmin of the *validation* grid, merely re-scored at that α on test. The Chronos runs share one fixed test subsample across seeds and one seeded zero-shot reference per dataset, since B–D divides by a shared denominator. Condition B was re-trained because only the encoder, not the head, had been checkpointed; its validation losses reproduce the published values to 0.008–0.039% (MPS run-to-run float noise), and all six frozen-encoder ridge optima reproduce exactly and remain interior.

Which cells count as degradation. The definition (gate-passing, for_B > 0 in every seed, for_D < 0 in every seed) selects exactly eight of the 31 cells, and `cell_matrix.py` applies it programmatically so the list cannot be curated by hand. The four TimesFM cells are put through the same test and none passes it. One further cell (Moirai-Small/ETTh2, $h=192$) meets the inequalities on *means* but its *forgetting%* term is only +2.0 with 5/10 seeds positive, so it is excluded and reported as directional.

Table 6 is emitted by `scripts/cell_matrix.py`, the same function that computes the body’s numbers, rather than transcribed, so it cannot drift out of step with Table 1. All dispersions here and in the body are SEM.

What moves. Condition B is stable across the two segments; condition D is not. On Chronos/ETTh1 the frozen head goes from 1.58/1.36/1.37 on validation to 1.81/2.51/2.07 on test, while B moves by a few percent. Chronos/ETTh2, the one cell that does not reverse, is also the one whose frozen head transfers cleanly (1.133 $\rightarrow$ 1.135). So the cell where freezing genuinely cost was robust to the choice of windows, and the cells where freezing appeared to *help* were the ones that did not survive it.

A consequence for the closed-form argument. We previously justified the ridge head as a closed-form optimum that admits no epoch budget. It is an optimum *on the validation windows*. Out of sample a weak validation-selected α transfers poorly, and the protocol-matched early-stopped AdamW head, which agrees with ridge to ~ 1 pp on validation, beats it on test (Chronos/ETTh1 seed 43: 2.505 against 1.957). We pre-committed to reporting the estimator validation preferred, which is ridge in every seed, with AdamW as a sensitivity; taking the better of the two on test would reintroduce exactly the fault this appendix removes. Under the AdamW leg the ETTh1 reversal is smaller but intact.

Table 6: Every cell, both scorings. Gate and B–D are computed on the same test windows; ^f marks the three cells that gate-fail there. “Reverses” = the sign of B–D differs between the two scorings.

Cell	Seeds	Gate R_{task}^2	B–D validation	B–D held-out	Reverses?
Chronos / ETTh1 ($h=24$)	3	+0.389	+6.8±2.9	− 39.2±13.9	yes
Moirai-S / ILI ($h=24$)	10	−0.033 ^f	−62.6±1.4	− 38.6±1.3	no
Chronos / ETTm2 ($h=24$)	3	+0.651	−24.1±9.8	− 29.5±6.4	no
Moirai-B / ETTh2 ($h=96, n=1000$)	3	+0.287	+3.8±4.1	− 22.3±2.9	yes
Chronos / Weather ($h=24$)	3	+0.494	−22.4±4.5	− 21.6±3.2	no
Moirai-L / ETTh2 ($h=96, n=500$)	3	+0.129 ^f	+7.2±5.8	− 20.8±1.4	yes
Moirai-B / ETTh2 ($h=192, n=1000$)	3	+0.554	−3.9±6.2	− 20.2±1.1	no
Chronos / ETTh2 ($h=24$)	3	+0.456	−19.9±3.2	− 12.9±0.6	no
TimesFM / ETTm2 ($h=24$)	3	+0.703	−12.2±12.4	− 7.7±2.6	no
TimesFM / ETTh2 ($h=24$)	3	+0.472	+0.2±2.9	− 7.1±3.1	yes
Moirai-S / Electricity7 ($h=96$)	3	+0.535	−10.4±1.1	− 3.2±1.5	no
Chronos / Electricity ($h=24$)	3	+0.138 ^f	−10.3±3.2	− 2.7±1.9	no
Moirai-S / Electricity7 ($h=192$)	3	+0.650	−8.2±1.9	− 2.6±2.1	no
TimesFM / Weather ($h=24$)	3	+0.506	−4.8±0.6	− 1.6±1.5	no
TimesFM / ETTh1 ($h=24$)	3	+0.401	−5.4±0.9	− 1.4±2.8	no
Moirai-B / ETTm2 ($h=192$)	3	+0.714	+3.5±1.5	+ 1.3±3.4	no
Moirai-S / ETTm2 ($h=96, n=1000$)	5	+0.568	+6.5±2.9	+ 2.6±3.4	no
Moirai-S / ETTh2 ($h=96, n=500$)	10	+0.331	+10.6±2.8	+ 2.8±3.7	no
Moirai-S / ETTm2 ($h=192, n=1000$)	5	+0.722	+8.6±3.3	+ 4.7±3.2	no
Moirai-S / ETTh2 ($h=192, n=500$)	10	+0.670	+17.4±4.3	+ 6.1±3.4	no
Moirai-B / ETTm2 ($h=96$)	3	+0.588	+3.9±3.7	+ 7.0±2.4	no
Moirai-B / Weather ($h=96$)	3	+0.485	+19.3±0.8	+ 9.4±2.0	no
Moirai-L / Weather ($h=96$)	3	+0.588	+4.3±6.5	+ 12.0±7.5	no
Moirai-B / Weather ($h=192$)	3	+0.752	+30.6±12.0	+ 12.5±8.7	no
Moirai-S / ETTh1 ($h=192, n=1000$)	5	+0.719	+14.8±1.9	+ 13.5±1.7	no
Moirai-S / ETTh1 ($h=96, n=1000$)	5	+0.578	+16.1±1.5	+ 22.1±2.3	no
Moirai-L / ETTh1 ($h=96$)	3	+0.506	+13.4±2.1	+ 28.2±4.9	no
Moirai-B / ETTh1 ($h=96$)	3	+0.505	+15.8±4.6	+ 37.5±8.2	no
Moirai-B / ETTh1 ($h=192$)	3	+0.682	+15.6±2.3	+ 47.0±13.8	no
Moirai-S / Weather ($h=96$)	3	+0.526	+55.4±4.3	+ 53.1±10.2	no
Moirai-S / Weather ($h=192$)	3	+0.825	+70.2±12.3	+ 62.1±13.6	no

595 **ILI.** This cell also required its condition D to be re-run from scratch, for a different reason: the
596 frozen control had never been trained (App. C). With a real D its validation B–D is -62.6 ± 1.4
597 and -38.6 ± 1.3 on held-out windows. Its D reaches CKA 0.76–0.90 rather than 1.000, because the
598 projections feeding the encoder stay trainable.

599 **Scope and denominators.** Every cell in Table 6 is scored on a chronologically disjoint test split. The
600 zero-shot test denominator matters more than it looks: B–D divides by a shared zero-shot term, and
601 each run stores one measured on *validation*, so using it against a test-window difference mixes scales;
602 on Moirai-Small/ETTh2 $h=96$ the test zero-shot is 0.492 against validation’s 0.129, which inflates
603 held-out B–D by $\approx 3.8\times$. The test references come from three run sets (results/v41_zs_test,
604 v43_moirai_matrix/*/condition_A, v39_moirai_zs_test), 13 in total, and the Chronos cells
605 share one fixed test subsample and one seeded zero-shot per dataset across seeds. Two rows of Table 1
606 in an earlier version, Chronos/M4 and the TimesFM $h=96$ pair, have no paired B/D artifacts with test
607 MSEs retained and are not among the 31; they appear only in App. E. The four TimesFM $h=24$ cells
608 are a different matter: they were run for this revision and scored on test windows from the outset, so
609 nothing about them was re-scored.

610 H The Strict-Freeze Control

611 Condition D freezes the encoder’s own weights but leaves the input projection (in_proj, 247,680
612 parameters) and mask_encoding (384) trainable alongside the distribution head (param_proj,
613 2,956,800). What the encoder *receives* therefore keeps changing, so its output is not a fixed function
614 of the input, which is why D’s measured CKA is 0.76–0.90 on ILI and 0.97–0.99 on Moirai-Base

rather than 1.000. A reader can reasonably object that “freezing wins” in the degradation cells is partly the input projection re-fitting rather than the encoder being held still.

Condition **H** removes the objection: `in_proj` and `mask_encoding` are frozen too, so only `param_proj` trains and the encoder’s output is pinned by construction (measured CKA 1.0000 exactly, as it must be). B–H is then a clean read of what allowing *any* encoder-side change was worth, and B–D versus B–H separates encoder adaptation from input re-fitting. We fixed the reading rule before running: the claim is that the sign and reading of B–D survive strict freezing, so B–H is reported beside B–D on every cell, and divergence on any cell is reported as that cell’s gap being partly input re-fitting rather than being buried.

Result. All six degradation cells known at the time, plus ILI, were run; nothing was dropped. The two the prospective arm later added (Moirai-Large on ETTh1 and on Weather) postdate this control and carry no H run, so the strict-freeze evidence covers six of the eight. Measured CKA is 1.0000 everywhere, confirming the construction. **The reading survives on every cell:** B–H keeps B–D’s sign in all seven, and the magnitude changes by -5.8 to $+0.5$ pp, so at most 5.8 pp of a gap as large as 62 pp is input-projection re-fitting. On ILI, where adaptation helps, strict freezing makes adaptation look *more* valuable ($-38.6 \rightarrow -43.6$), which is the same direction: holding more of the model still costs more where the encoder was earning its keep. B–D is averaged over the seeds condition H ran, so Moirai-Small/ETTh1’s entries here are 3-seed values against the 5-seed values quoted in Table 1.

Cell	Seeds	B–D	B–H	for _H	CKA _H
Moirai-S / Weather ($h=192$)	3	$+62.1 \pm 13.6$	$+56.7 \pm 13.5$	-10.6 ± 0.3	1.0000
Moirai-S / Weather ($h=96$)	3	$+53.1 \pm 10.2$	$+50.1 \pm 6.6$	-12.8 ± 0.6	1.0000
Moirai-B / ETTh1 ($h=192$)	3	$+47.0 \pm 13.8$	$+47.5 \pm 13.7$	-8.0 ± 0.4	1.0000
Moirai-B / ETTh1 ($h=96$)	3	$+37.5 \pm 8.2$	$+32.5 \pm 8.2$	-9.7 ± 0.4	1.0000
Moirai-S / ETTh1 ($h=96, n=1000$)	3	$+23.2 \pm 3.6$	$+20.4 \pm 3.6$	-0.6 ± 0.2	1.0000
Moirai-S / ETTh1 ($h=192, n=1000$)	3	$+14.5 \pm 1.4$	$+8.7 \pm 1.5$	$+1.2 \pm 0.4$	1.0000
Moirai-S / ILI ($h=24$)	10	-38.6 ± 1.3	-43.6 ± 0.7	-7.1 ± 0.5	1.0000

One clause does not survive, and it is not the headline one. “Degradation” has two parts: fine-tuning ends up worse than the un-tuned model, *and* freezing improves on it. The first part is a property of condition B and is untouched by this control. The second is condition D’s, and on Moirai-Small/ETTh1 $h=192$ the strictly-frozen model does *not* beat the un-tuned model: $\text{for}_H = +1.2 \pm 0.4\%$, positive in all 3 seeds, where for_D is $-4.6 \pm 0.6\%$. On that cell the head alone cannot recover the zero-shot forecast; D’s improvement came partly from re-fitting the input projection. The other five cells keep both parts (for_H from -0.6% to -12.8%), and B–H stays positive on all six, so “freezing beats full fine-tuning” holds throughout while “freezing also beats not fine-tuning” holds strictly on five of six. We report the cell rather than re-defining the term around it.

I CKA Pooling Robustness

A consequence worth stating plainly: because the magnitude of CKA depends on how the representation is constructed, this paper’s claims are strongest where they are *ordinal and interventional* (which cells drift, and what freezing costs on each), and weakest if read as statements about absolute CKA values. The drift *flag* is stable under pooling; the number attached to it is not.

Pooling robustness. CKA is reported on mean-pooled encoder outputs throughout. Because mean-pooling discards token-level structure, we recomputed whole-encoder CKA on the same ten fine-tuned ETTh2 encoders ($n=10k, h=96$, condition B) both mean-pooled and token-flattened, over 500 held-out 192-timestep windows.

Representation	CKA (mean $\pm$ SD, 10 seeds)	Seed range
Mean-pooled (as reported)	0.561 ± 0.157	0.328–0.840
Token-flattened	0.732 ± 0.036	0.685–0.797

Two things follow. The *qualitative* drift flag is robust: both values sit far below the 0.95 threshold, so every cell keeps its assignment. The *magnitude* is not: token-level similarity is substantially higher

and far less seed-variable (± 0.036 vs ± 0.157), so part of the drift reported as ~ 0.52 is an artifact of pooling. We regard this as consistent with, rather than damaging to, the thesis (it is a further reason not to read a single scalar as a measure of damage), but CKA magnitudes should be read as pooling-dependent.

J Predicting the Reading Without the Frozen Control

Within Moirai the reading is recoverable from one run. Condition B’s *own* held-out forgetting agrees in sign with B–D in 16 of the 22 Moirai cells, and the matched within-checkpoint contrast is sharper: Moirai-Base forgets under B on ETTh1 at both horizons (+22.8%, +39.5%) and favours freezing, while the *same* checkpoint on ETTh2 improves under B (−31.4%) and strongly favours adaptation. One fine-tuning run therefore distinguishes regimes CKA does not ($\rho = +0.894$ vs CKA’s $+0.168$, $n=22$).

We state the coupling rather than bank it. $B-D \equiv \text{forg}_B - \text{forg}_D$ identically, and forg_D barely moves (SD 8.1 against forg_B ’s 26.2), so the rule works because the frozen arm is nearly constant, an empirical fact about Moirai, not a theorem. It fails on Chronos ($\rho = -0.100$, limitation iii), and on TimesFM, the other arm where forg_B is a clean measurement, the sign agrees 4 of 4 while forg_D varies nearly as much as forg_B (SD 19.2 vs 22.0): the mechanism does not transfer even where the rule does. It is reported here, not in the body, because it is a fourth result competing with the three the paper is actually about, and because a rule that holds on one backbone for a reason that does not generalise is not a contribution we want weighed alongside them.

K Related Work

Catastrophic-forgetting mitigations such as EWC [Kirkpatrick et al., 2017], LwF [Li and Hoiem, 2018], LoRA [Hu et al., 2022] and L2-SP [Li et al., 2018] are designed to limit how far a fine-tuned model moves from its pretrained state, but they are typically applied without first asking whether the pretrained state is worth preserving on the target task. Our pretrained-capability inclusion criterion is a downstream-only check for exactly that question, and the preservation spectrum (Appendix D) shows why it matters: methods that raise representational similarity do not reliably improve utility, and the ordering reverses with horizon. The reflex itself is well founded where it originated: Luo et al. [2023] document substantial forgetting in LLMs under continual fine-tuning. Our claim is not that the phenomenon is absent but that on TSFMs the drift measurement used to detect it does not order downstream damage.

Closest to this work, Wiliński et al. [2025] study TSFM representations directly: they measure layer self-similarity within and across model sizes, find block-like redundancy that supports informed pruning, and steer the latent space to inject concepts such as periodicity and trend. Their interventions act on a *trained model’s activations* to characterize and control what the representation encodes; ours acts on the *learning process* (freezing the encoder and re-running fine-tuning) to ask whether representation change was necessary for the downstream gain. The two are complementary, and their redundancy finding is a plausible mechanism for what we observe: if TSFM representations carry substantial redundancy, a large measured drift can be reorganization within that slack rather than loss of task-relevant structure, which is exactly the gain-independent restructuring our intervention brings out. Neither their analysis nor ours predicts a cell’s B–D without running both conditions.

Kumar et al. [2022] show that fine-tuning can distort pretrained features and that linear probing followed by fine-tuning mitigates it; Neyshabur et al. [2020] study what transfers between pretrained and fine-tuned networks. Neither provides a pre-fine-tuning screen for when preservation is warranted, which is the gap this paper addresses. On the measurement side we use linear CKA [Kornblith et al., 2019]; our pooling analysis (Appendix I) is a reminder that its magnitude is representation-dependent and should be read as an operational drift detector rather than a measure of damage.

Pretraining overlap for the backbones we test is reported in Appendix E.

L Moirai-Base Frozen-Encoder Run

The Moirai-Base/ETTh2 cell is the most heavily drifted cell in the study but originally had no frozen-encoder control, so nothing was known about what allowing encoder adaptation was worth there. Two comparisons are available, and we lead with the cleaner one.

A matched, paired comparison at $n=1000$. Here both conditions run on the *same* three seeds and the *same* hardware (20 epochs, no early stopping), so the contrast is internally controlled.

Seed	B (full FT)	D (frozen)	B−D
42	−35.33	−41.64	+6.31
123	−47.93	−43.62	−4.31
456	−47.57	−56.88	+9.31
Mean	−43.61	−47.38	+3.77

Freezing the encoder is if anything *better* here, in 2 of 3 seeds: on Moirai-Base, encoder adaptation does not account for the gain.

A larger estimate at $n=10k$, across hardware. We also ran condition D under the published condition-B protocol (Moirai-Base, ETTh2, $h=96$, $n=10k$, $lr 10^{-4}$, early stopping) on the same ten seeds as the published condition B, on MPS.

Seed	Forgetting%	Final CKA	Weight drift	Best epoch
42	−49.33	0.984	1.88	7
101	−54.13	0.972	2.87	16
123	−45.85	0.985	1.90	6
202	−40.14	0.983	1.77	6
303	−43.27	0.990	1.44	4
456	−40.71	0.979	2.58	12
777	−38.03	0.977	2.15	9
789	−49.86	0.983	2.13	8
888	−41.25	0.969	2.53	10
999	−50.71	0.976	2.50	11
Mean±std	−45.33±5.43	0.980±0.006		

All 10 seeds are negative (SEM 1.72). Against the published condition B ($-54.2\pm 8.6\%$, 10 seeds) this is a B−D gap of 8.9 pp: a modest incremental value for allowing encoder adaptation, despite CKA falling to 0.25–0.33 under full fine-tuning, the deepest drift anywhere in the study. This estimate is **not internally controlled**: the published B ran on CUDA and this D on MPS. At ten seeds each the standard deviations still overlap, but the means no longer do (Welch $t=2.8$ on the 8.9 pp difference), so the 8.9 pp figure cannot be separated from the hardware change, and the matched pair above carries the conclusion. Earlier partial runs are excluded throughout.

The 10-seed run also shows that condition D is not representation-frozen. Early stopping selected epochs 6–16 rather than epoch 0, and the frozen-encoder CKA settles at 0.97–0.98 with non-zero weight drift (1.88–2.87), because the input/output projections outside the encoder remain trainable. In cells where early stopping selects the untrained state (ILI, Chronos/M4) the measured CKA is 1.000, which earlier drafts read as holding by construction.

M Threshold Sensitivity

Both thresholds are operational decision boundaries for this study, not universal constants, and neither was tuned to produce the reported signatures.

Preservation-priority gate (20%), and a margin claim we withdraw. Earlier versions reported the observed gate values as bimodal with a wide margin: “no cell falls in $[0, 28)\%$, so any threshold in 10–30% produces identical assignments”. That was read on the windows model selection uses. Re-computed on the test windows for all 31 cells from one code path (`scripts/gate_all_cells.py`, §3), the values are −3.3, 12.9, 13.8, 28.7, 33.1, 38.9, 40.1, 45.6, 47.2, 48.5, 49.4, 50.5, 50.6, 50.6,

52.6, 53.5, 55.4, 56.8, 57.8, 58.8, 58.8, 65.0, 65.1, 67.0, 68.2, 70.3, 71.4, 71.9, 72.2, 75.2, 82.5% (the four TimesFM cells contribute 40.1, 47.2, 50.6 and 70.3). Three cells fall below the threshold (-3.3 , 12.9 , 13.8) and one sits just above it at 28.7% , so the distribution is not bimodal out of sample and the threshold is *not* margin-free: moving it to 30% would additionally reclassify Moirai-Base/ETTh2 $h=96$, and moving it to 10% would restore Chronos/Electricity and Moirai-Large/ETTh2 to gate-pass. The eight degradation cells are unaffected across the whole $10\text{--}50\%$ range (they sit at $50.5\text{--}82.5\%$), which is the assignment the paper’s claims depend on; the reclassified cells all read *adapt*, where the gate is not what licenses the reading.

CKA threshold reclassification. The gate analysis above shows wide margins; the CKA cutoff deserves the same treatment. We reclassify every cell at four thresholds using the observed CKA values (ranges span seeds, not datasets).

Thresh.	Moirai-B (0.25–0.33)	Moirai/ILI (0.456)	Chronos/ETT (0.087–0.179)	TimesFM ($h=24$) (0.19–0.64)	Chronos/M4 (0.98–0.999)	TimesFM ($h=96$) (0.995–0.999)
0.900	drift	drift	drift	drift	stable	stable
0.925	drift	drift	drift	drift	stable	stable
0.950	drift	drift	drift	drift	stable	stable
0.975	drift	drift	drift	drift	stable	stable

Every assignment is identical at all four thresholds, and no cell splits. Note that TimesFM appears twice and on opposite sides: its $h=96$ cells are the most stable in the study and its $h=24$ cells among the most drifted, so “stable” is not a property of the backbone. The margin is not symmetric, and it is worth being explicit about which end is tight: the drifting cells sit at least 0.26 below the lowest cutoff considered, whereas the stable cells clear the highest one by only 0.005 (Chronos/M4, whose lowest seed is 0.98) and 0.020 (TimesFM at $h=96$). So it is the stable end, not the drifting end, that would become threshold-sensitive if the cutoff were pushed past 0.975 . We state the operating range explicitly rather than claiming threshold-independence: the conclusions hold for cutoffs in $[0.90, 0.975]$.

The measured Chronos/ETT values sit an order of magnitude below the 0.95 cutoff, so nothing is *split* at that end.

CKA (0.95). Observed CKA clusters away from the cutoff at the drifting end: drifting cells sit at $0.087\text{--}0.179$, $0.19\text{--}0.64$, $0.25\text{--}0.33$, and 0.456 ; stable cells at $0.98\text{--}0.999$. A random-initialization baseline gives CKA 0.000 ± 0.000 (5 seeds), fixing the floor of the scale, so the Chronos/ETT cells at $0.087\text{--}0.179$ are much nearer that floor than the threshold. This is also why the Chronos/ETTh1-vs-ETTh2 comparison cannot be resolved by CKA at any cutoff: both cells are unambiguously drifted on every reading, yet the less-drifted of the two is the one that loses more to freezing.